Test Series Code

07

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | DROPPER | **M. MARKS: 720**

DATE: 08/03/2026

SYLLABUS

Physics:	Moving Charges and Magnetism, Magnetism and Matter, Electromagnetic Induction + 30% Questions from cumulative syllabus of previously conducted all AITS test
Chemistry:	Aldehydes, Ketones and Carboxylic Acids, Amines + 30% Questions from cumulative syllabus of previously conducted all AITS test
Botany:	Respiration in Plants + 30% Questions from cumulative syllabus of previously conducted all AITS test
Zoology:	Evolution + 30% Questions from cumulative syllabus of previously conducted all AITS test

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. The current sensitivity of a moving coil galvanometer is increased by 100% by increasing the number of turns. If the rest of the setup remains unchanged, what will be the percentage change in its voltage sensitivity?

(1) Increases by 100%
(2) Decreases by 50%
(3) 0% change
(4) Increases by 200%

2. What is the dimensional formula of the torsion constant (k) used in a moving coil galvanometer?

(1) $[ML T^{-2}]$ (2) $[MT^{-2}]$
(3) $[ML^2T^{-2}]$ (4) $[ML^{-1}T^{-2}]$

3. A 100-turn closely wound circular coil of radius 10 cm carries a current of $\frac{10}{\pi}$ A. The coil is placed at rest in a horizontal magnetic field of 2T with its plane initially parallel to the field. If the moment of inertia of the coil is 0.1 kg m^2, what is the angular speed acquired by the coil when it rotates by 90°?

(1) 10 rad/s (2) 20 rad/s
(3) 30 rad/s (4) 40 rad/s

4. Three long parallel wires are arranged as shown in the figure. All wires carry a current of equal magnitude. $(\otimes)$ represent current into the page, $(\odot)$ represents current out of the page. If Wire X is equidistant from the two wires, what is the direction of the net magnetic force on Wire X?

$\otimes$ $\odot$

X $\otimes$

(1) Towards the right ($\rightarrow$)
(2) Vertically downwards ($\downarrow$)
(3) Vertically upwards ($\uparrow$)
(4) Towards the left ($\leftarrow$)

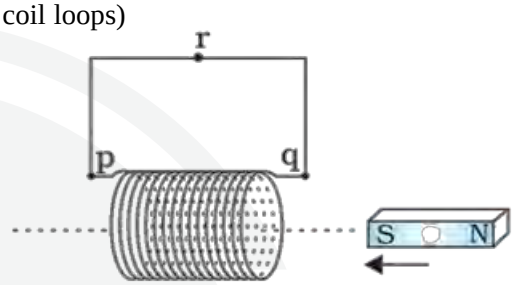
5. In a ferromagnetic material, individual atoms align themselves in macroscopic volumes called domains. What is the approximate size of a domain, and how many atoms does it contain?

(1) Size ≈ 1 cm; Atoms $\approx 10^{23}$
(2) Size ≈ 1 mm; Atoms $\approx 10^{11}$
(3) Size ≈ 1 mm; Atoms $\approx 10^{6}$
(4) Size ≈ 10 mm; Atoms $\approx 10^{15}$

6. A circular coil of 50 turns with an area of 0.2 m^2 is rotating in a uniform magnetic field of 1.0 T at a constant angular velocity of 2 rad/s. What is the instantaneous induced EMF when the plane of the coil makes an angle of 60° with the magnetic field lines?

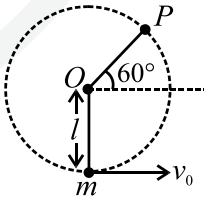
(1) 10 V (2) $10\sqrt{3}$ V
(3) 20 V (4) 50 V

7. A bar magnet is moved towards a solenoid as shown in the figure, with its South Pole (S) facing the coil. What will be the direction of the induced current in the external wire segment containing point r? (wire at terminal q comes from the front face of the coil loops)



(1) From $p \rightarrow r \rightarrow q$
(2) From $q \rightarrow r \rightarrow p$
(3) No current will be induced
(4) The direction oscillates continuously

8. A bob of mass m is suspended by a light string of length l. The bob is given a horizontal velocity v_0 at the bottommost point. The string becomes slack at a point P where the string makes an angle of 60° with the horizontal as shown in the figure.



What is the ratio of the speed v of the bob at point P to its initial speed v_0?

(1) $\sqrt{\frac{1}{4+3\sqrt{3}}}$ (2) $\sqrt{\frac{\sqrt{3}}{2+3\sqrt{3}}}$
(3) $\sqrt{\frac{\sqrt{3}}{4+3\sqrt{3}}}$ (4) $\sqrt{\frac{3}{4+2\sqrt{3}}}$

9. An electron is moving in a circular path under the influence of a transverse magnetic field of 3.57×10^{-2} T. If the value of $\frac{e}{m}$ is 1.76×10^{11} C/kg, the frequency of revolution of the electron is:

(1) 100 MHz (2) 1 GHz
(3) 62.8 MHz (4) 6.28 MHz

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10. A wire of length L is bent into a circular coil of N turns. If a current I is passed through it, the magnetic field at the center is B. If the same wire is bent into a circular coil of n turns and the same current is passed, the new magnetic field at the center will be:

(1) $\frac{n}{N}B$ (2) $\frac{n^2}{N^2}B$

(3) $\frac{N^2}{n^2}B$ (4) n^2B

11. A circular loop of radius R carries current I. Let the magnetic field at the center be B_c and the magnetic field at a point on the axis at distance $x=\sqrt{3}R$ be B_a. The ratio $B_c:B_a$ is:

(1) 2 : 1 (2) 4 : 1

(3) 8 : 1 (4) 16 : 1

12. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): When a magnet is dropped through a long vertical copper tube, it eventually achieves a constant terminal velocity.

Reason (R): The eddy currents generated in the walls of the tube produce a magnetic field that exerts a retarding force, which eventually becomes equal to the weight of the magnet.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

13. A magnet of magnetic moment M is suspended in a uniform magnetic field B. The time period of small oscillations is T. If the magnet is cut into two equal halves perpendicular to its length and one piece is suspended in the same field, the new time period will be:

(1) $\frac{T}{2}$ (2) T

(3) $2T$ (4) $\frac{T}{\sqrt{2}}$

14. A long, straight cylindrical wire of radius R carries a steady current I which is uniformly distributed over its cross-section. Let B_1 be the magnetic field at a distance $\frac{R}{3}$ from the axis of the wire, and B_2 be the magnetic field at a distance 3R from the axis. The ratio $B_1:B_2$ is:

(1) 1 : 1 (2) 3 : 1

(3) 9 : 1 (4) 1 : 3

15. A conducting ring of radius r is placed perpendicularly inside a time-varying magnetic field given by $B=B_0+\alpha t$. The electric field induced at the circumference of the ring is: (B_0 is a positive constant)

(1) $\frac{\alpha r}{2}$ (2) αr

(3) $2\alpha r$ (4) $\frac{\alpha r^2}{2}$

16. Considering rules of significant figures, which one of the following operations is **correct**?

(1) $10.20\text{ cm}+9.00\text{ cm}=19.2\text{ cm}$

(2) $2.54\text{ m}-1.2\text{ m}=1.34\text{ m}$

(3) $4.1\text{ m}\times1.4\text{ m}=5.74\text{ m}^2$

(4) $3.6\text{ m}/1.75\text{ s}=2.1\text{ m/s}$

17. A magnetic needle lying parallel to a magnetic field requires W units of work to turn it through 60°. The torque needed to maintain the needle in this position will be:

(1) $\sqrt{3}W$ (2) W

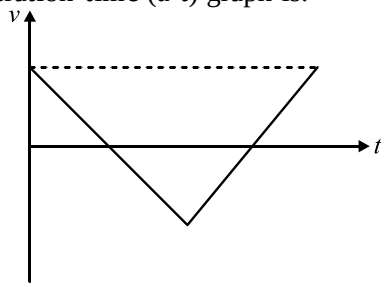
(3) $\frac{\sqrt{3}}{2}W$ (4) 2W

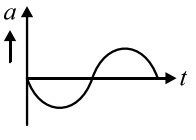
18. A wire with a linear mass density of (2π)g/m carries a current of 10 A and wire is placed horizontally and floats perpendicular to the axis inside a long horizontal solenoid (500 turns/m). The current required in the solenoid windings to support the wire's weight is: (Take $g=10\text{ m/s}^2$)

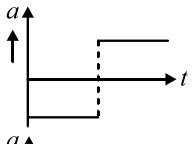
(1) 100 A (2) 10 A

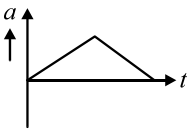
(3) 1 A (4) 50 A

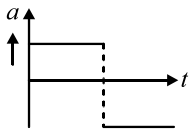
19. The velocity-time (v-t) graph of a particle moving in straight line is shown in figure. The corresponding acceleration-time (a-t) graph is:



(1) 

(2) 

(3) 

(4) 

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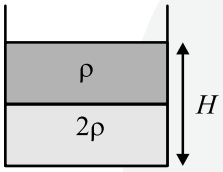
20. If a diamagnetic liquid is placed in a U-tube and one arm is placed between the poles of a strong magnet, the liquid level in that arm will:

(1) Rise (2) Fall
(3) Remain same (4) Oscillate

21. A long solenoid carrying a constant current produces a magnetic field B_0 in its interior (vacuum). When a core of a specific ferromagnetic material is inserted completely into the solenoid, the magnetic field inside the solenoid increases to $500B_0$. What is the value of the magnetic susceptibility (χ) of the core material?

(1) 500 (2) 499
(3) 501 (4) 999

22. Two immiscible, incompressible liquids with densities ρ and 2ρ are filled in a cylindrical container to a total height of *H* which has base area *A*. Both the liquids have equal masses. What is the force exerted on the base of the container due to liquid? (ignore atmospheric pressure)

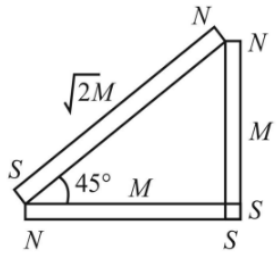


(1) $\frac{4\rho gHA}{3}$ (2) $\frac{3\rho gHA}{4}$
(3) $\frac{5\rho gHA}{3}$ (4) $\frac{\rho gHA}{2}$

23. A short bar magnet has a magnetic moment of $0.4\ \text{Am}^2$. The magnitude of the magnetic field at a distance of 20 cm from the centre of the magnet on its equatorial line (broad-side on position) is:

(1) 5×10^{-6} T
(2) 10×10^{-6} T
(3) 5×10^{-5} T
(4) 1×10^{-7} T

24. The magnetic moment of the arrangement shown in the figure is:



(1) Zero (2) $2\sqrt{2}M$
(3) $2M$ (4) M

25. An inductor ($L = 4$ H) and a resistor ($R = 10\ \Omega$) are connected in series with a 20 V battery. The switch is closed at $t = 0$. At what time t will the potential difference across the inductor (V_L) be equal to the potential difference across the resistor (V_R)? (Use $\ln 2 \approx 0.69$)

(1) 0.28 s (2) 0.40 s
(3) 0.69 s (4) 1.38 s

26. Maximum possible mutual inductance due to coupling of inductors having self inductances L_1 and L_2 is:

(1) $L_1 + L_2$ (2) L_1L_2
(3) $\sqrt{L_1L_2}$ (4) $\frac{L_1}{L_2}$

27. Which of the following option contains **incorrect** statements for Lenz's law of electromagnetic induction?

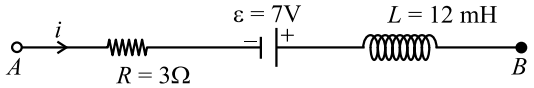
A. The induced EMF produced is such that it supports the change in magnetic flux.
B. The induced current flows in a direction such that it opposes the change that caused it.
C. When the magnetic flux through a coil changes rapidly, the magnitude of the induced EMF is smaller.
D. The value of induced current passing through a circuit depends on the total change in magnetic flux only.

(1) A and D only
(2) B, C and D only
(3) A, C and D only
(4) A and C only

28. A closed circular coil has 50 turns and a total resistance of 10 Ω. If the magnetic flux linked to each turn changes by 2×10^{-2} Wb in 0.1 seconds, what is the total charge that flows through the coil?

(1) 0.01 C (2) 0.1 C
(3) 1.0 C (4) 10 C

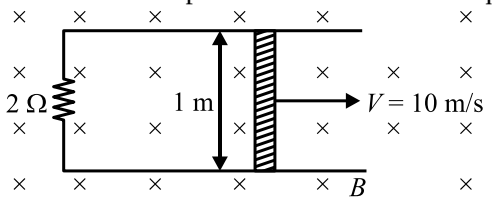
29. The electric network shown in figure is part of a complete circuit. If at any instant, current $i = 2$A and potential difference across A and B is zero, then value of $\left|\frac{di}{dt}\right|$ is:



(1) $\frac{26}{3}$ A/s (2) $\frac{25}{3}$ A/s
(3) $\frac{250}{3}$ A/s (4) $\frac{290}{3}$ A/s

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30. A metal rod slides on conducting rails as shown in figure. The circuit area is perpendicular to magnetic field with $B = 0.4\text{T}$. If the resistance of circuit is 2Ω, the force needed to move the rod as indicated with a constant speed of 10 m/s will be equal to:

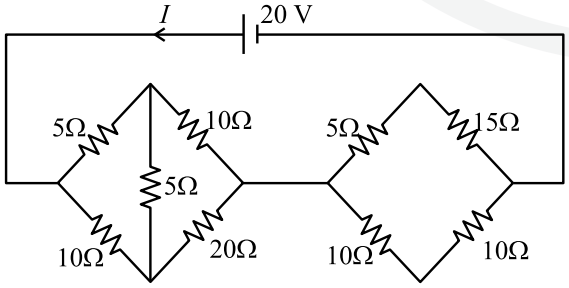


(1) 8×10^{-4} N
(2) 8×10^{-2} N
(3) 8×10^{-1} N
(4) 8×10^{-3} N

31. Two tuning forks of frequencies 512 Hz and 526 Hz respectively are vibrated simultaneously, then:
(1) Beat frequency heard by a person will be 14 Hz
(2) Beat frequency heard by a person will be 10 Hz
(3) Beats will not be produced
(4) Beats will produce but not heard by a person

32. A charged particle with specific charge $\frac{q}{m}$ equal to 10^8 C/kg enters a uniform magnetic field $B = 2\pi\times10^{-2}$ T with a velocity $v = 2\times10^4$ m/s at an angle of 60° to the direction of the field. The pitch of the helical path is:
(1) 1 cm (2) 2 cm
(3) 50 cm (4) 0.5 cm

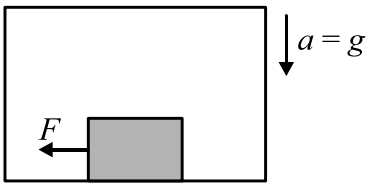
33. The value of current I in the circuit is:



(1) 0.5 A (2) 1 A
(3) 1.5 A (4) 2 A

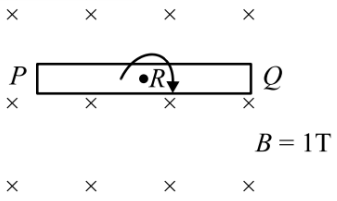
34. Given below are two statements:
Statement I: The bulk modulus of elasticity for gas depends on the nature of gas (monoatomic, diatomic, etc) for adiabatic process.
Statement II: Bulk modulus of elasticity is zero for isobaric process.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

35. A block of mass m is kept on the rough floor of a freely falling lift. During the free fall of the lift, the block is pulled horizontally with a force of F = 5 N, if coefficient of friction between block and lift is $\mu_s = 0.1$, then frictional force on the block will be:



(1) 5 N (2) 2 N
(3) zero (4) 10 N

36. A rod PQ of length 30 cm fixed at its centre R rotates clockwise in a plane perpendicular to the field with a uniform angular velocity 10 rad s^{-1} about its perpendicular bisector in a uniform magnetic field of 1 T.

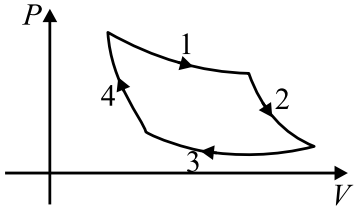


Which of the following statements is **correct** regarding the potentials at ends P, Q and the centre R?
(1) $V_P - V_Q = 0.225\,\text{V}$
(2) $V_Q - V_R = 0.3125\,\text{V}$
(3) $V_P = V_Q$, and R is the point of lowest potential.
(4) $V_P = V_Q$, and R is the point of highest potential.

37. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Gauss's law can be used to calculate electric field due to charged conducting spherical shell.
Reason R: Charged conducting spherical shell has symmetrical charge distribution.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false
(2) A is false but R is true
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

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38. The graph given below represents Carnot cycle of a heat engine. Match the processes 1, 2, 3, 4 in **List-I** with their characteristics given in **List-II**. (symbols have usual meaning)



List-I (Process)		List-II (Characteristics)	
(A)	1	(I)	Adiabatic expansion
(B)	2	(II)	Isothermal compression
(C)	3	(III)	Isothermal expansion
(D)	4	(IV)	Adiabatic compression

(1) A-I, B-II, C-III, D-IV
(2) A-III, B-I, C-II, D-IV
(3) A-IV, B-III, C-I, D-II
(4) A-I, B-III, C-IV, D-II

39. Two identical satellites *A* and *B* revolve round the earth in circular orbits at distances *R* and 3*R* from the surface of the earth (*R* = radius of the earth). The ratio of the angular momenta of *A* to that of *B* is:

(1) 1 : 1 (2) $1:\sqrt{2}$
(3) $\sqrt{2}:1$ (4) 2 : 1

40. Two small magnets, each having a magnetic moment of 5 Am^2, are placed in an end-on position (along the same axis) such that the distance between their centres is 0.1 m. The magnitude of the magnetic force acting between them is:

(1) 1.5 N
(2) 0.15 N
(3) 1.5×10^{-3} N
(4) 0.015 N

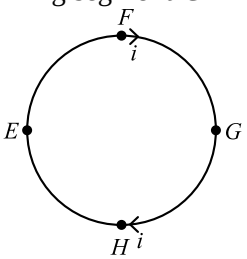
41. A magnetising field of 1600 A/m produces a magnetic flux of 2.4×10^{-5} Wb in an iron bar of cross-sectional area 0.2 cm^2. The magnetic susceptibility of the iron bar is: (Take $\mu_0 = 4\pi\times10^{-7}$ T m/A):

(1) 596 (2) 298
(3) 1192 (4) 1600

42. An uncharged capacitor of capacitance 4 μF is connected with a battery of 10 volt. The energy produced as heat energy in connecting wires till the capacitor is completely charged would be:

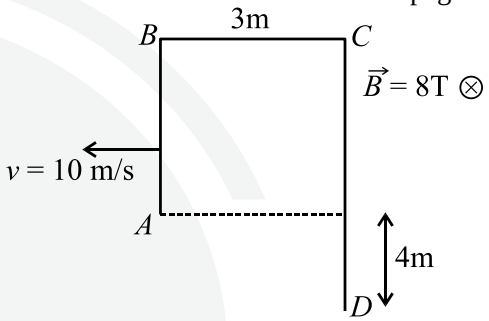
(1) 200 μJ (2) 300 μJ
(3) 600 μJ (4) 100 μJ

43. A circular coil *EFGH* carrying a current *i* is placed perpendicular to field in a uniform magnetic field. If magnetic force on the segment *EFG* is $\vec{T}$, then force on the remaining segment *GHE* is:



(1) $2\vec{T}$ (2) $\vec{T}$
(3) $-\vec{T}$ (4) $-2\vec{T}$

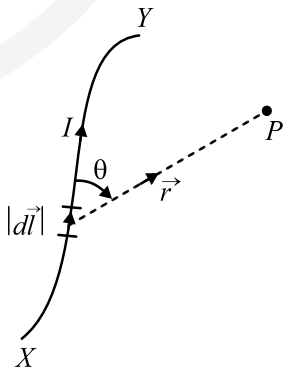
44. A metallic wire, bent into a shape *ABCD* shown below, moves to the left with a constant velocity of 10 m/s. It is traveling through a uniform magnetic field of 8 T that is directed into the page.



What is the magnitude of the electromotive force induced between the ends *A* and *D*?

(1) 400 V (2) 320 V
(3) 880 V (4) 0 V

45. A finite conducting wire *XY* carries a current *I*. The magnetic field $d\vec{B}$ due to infinitesimal element $d\vec{l}$ of the conducting wire at a point *P* as shown in figure is:



(1) $d\vec{B} = \frac{\mu_0}{4\pi} i^2 \left(\frac{d\vec{l} \times \vec{r}}{r} \right)$

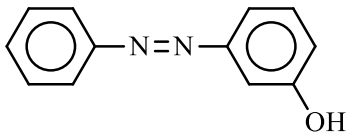
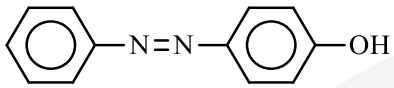
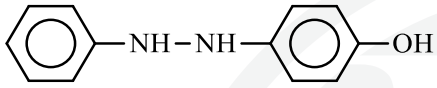
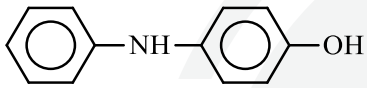
(2) $d\vec{B} = \frac{\mu_0}{4\pi} i \left(\frac{d\vec{l} \times \vec{r}}{r^3} \right)$

(3) $d\vec{B} = \frac{\mu_0}{4\pi} i \left(\frac{d\vec{l} \times \vec{r}}{r} \right)$

(4) $d\vec{B} = \frac{\mu_0}{4\pi} i^2 \left(\frac{d\vec{l} \times \vec{r}}{r^2} \right)$

[Contd...

46. CH_3CH_2OH can be prepared by the reaction of:
(1) $HCHO + CH_3MgCl$ followed by hydrolysis
(2) $CH_3CHO + CH_3MgCl$ followed by hydrolysis
(3) $HCHO + CH_3CH_2MgCl$ followed by hydrolysis
(4) $CO_2 + CH_3MgCl$ followed by hydrolysis

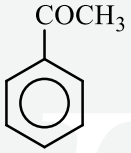
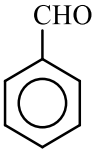
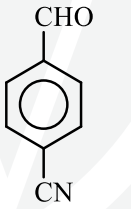
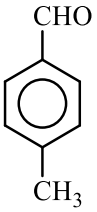
47. Benzene diazonium chloride on reaction with phenol in basic medium gives:
(1) 
(2) 
(3) 
(4) 

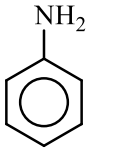
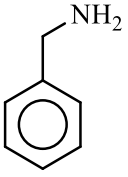
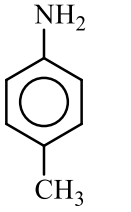
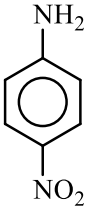
48. The following reaction is an example of: $CH_3CH_2Cl \xrightarrow[\Delta]{\text{alc.KOH}} CH_2=CH_2$
(1) Elimination reaction
(2) Substitution reaction
(3) Rearrangement reaction
(4) Addition reaction

49. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: $NaHSO_3$ is used for separation and purification of aldehydes.
Reason R: Hydrogen sulphite addition compound of aldehydes are water insoluble.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

50. Given below are two statements:
Statement I: Isocyanide test is used as a test for primary amines.
Statement II: Aniline does not undergo Friedel-crafts alkyation reaction.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

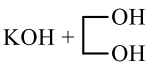
51. $\Delta_{sub}H^o(Li), \Delta_iH^o(Li), \Delta_{BE}H^o(F_2)$, $\Delta_{lattice}H^o(LiF)$ and $\Delta_fH^o(LiF)$ are 161, 520, 154, –1047 and –617 kJ mol^{-1} respectively. The $\Delta_{eg}H^o(F)$ will be:
(1) –132 kJ mol^{-1}
(2) –328 kJ mol^{-1}
(3) –220 kJ mol^{-1}
(4) –420 kJ mol^{-1}

52. Which among the following compounds is most reactive towards nucleophilic addition reaction?
(1) 
(2) 
(3) 
(4) 

53. The most basic compound among the following is:
(1) 
(2) 
(3) 
(4) 

54. Number of moles of KI oxidised by one mole of $KMnO_4$ in faintly alkaline medium is:
(1) 3
(2) 1
(3) 0.5
(4) 2.5

55. Match **List-I** with **List-II**.

List-I (Reaction)		List-II (Reagent)	
(A)	Clemmensen reduction	(I)	$SnCl_2$, HCl
(B)	Etard reaction	(II)	Zn–Hg, HCl
(C)	Wolff-Kishner reduction	(III)	CrO_2Cl_2 in CS_2
(D)	Stephen reaction	(IV)	(i) N_2H_4 (ii) KOH + 

Choose the **correct** answer from the options given below:

(1) A-III; B-IV; C-II; D-I
(2) A-II; B-IV; C-I; D-III
(3) A-IV; B-I; C-II; D-III
(4) A-II; B-III; C-IV; D-I

56. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: $CH_3–NH–C_2H_5$ is a secondary amine.
Reason R: It is a mixed amine.
In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

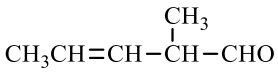
57. Total number of nodes present in 4p orbital is:

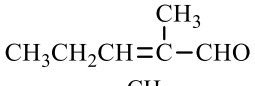
(1) 1 (2) 2
(3) 3 (4) 4

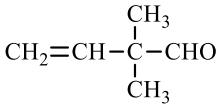
58. Identify the major product of the following reaction.

$CH_3CH_2CHO \xrightarrow[\Delta]{\text{dil. NaOH}}$

(1) $CH_3CH_2CH{=}CHCH_2CHO$

(2) 

(3) 

(4) 

59. During the nitration of aniline in strong acidic medium, percentage of ortho, meta and para derivatives obtained among its nitro derivative at 288K respectively are:

(1) 47%, 2% and 51%
(2) 51%, 2% and 47%
(3) 2%, 47% and 51%
(4) 2%, 51% and 47%

60. Given below are two statements:

Statement I: The conjugate acid of SO_4^{2-} is HSO_4^-.

Statement II: Aqueous solution of NH_4Cl is acidic in nature due to anionic hydrolysis.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

61. Most acidic compound among the following is:

(1) C_6H_5COOH
(2) FCH_2COOH
(3) HCOOH
(4) $ClCH_2CH_2COOH$

62. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Among isomeric amines, 3°-amines have minimum boiling point.
Reason R: 3°-amines do not have intermolecular association due to absence of hydrogen atom available for H-bond formation.
In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

63. Match **List-I** with **List-II**.

List-I (Sample)		List-II (Number of atoms)	
(A)	22 g CO_2	(I)	N_A
(B)	32 g O_2	(II)	0.75 N_A
(C)	16 g SO_2	(III)	2 N_A
(D)	4 g He	(IV)	1.5 N_A

Choose the **correct** answer from the options given below:

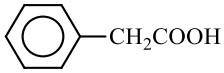
(1) A-III; B-IV; C-II; D-I
(2) A-IV; B-III; C-II; D-I
(3) A-IV; B-I; C-II; D-III
(4) A-II; B-III; C-IV; D-I

[Contd...

64. Which of the following compounds will give HVZ reaction?

(A) CH_3CH_2COOH

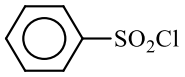
(B) CH_3COOH

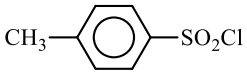
(C) 

(1) A and B only (2) B and C only

(3) A and C only (4) A, B and C

65. Which of the following is/are used for the identification of 1°, 2° and 3° amines?

(A) 

(B) 

(C) $CH_3–SO_2Cl$

(1) A only (2) A and B

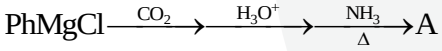
(3) A and C (4) B and C

66. Which of the following alkanes will **not** be obtain in good yield by Wurtz reaction?

(1) n-Butane (2) Ethane

(3) Propane (4) 2,3-Dimethylbutane

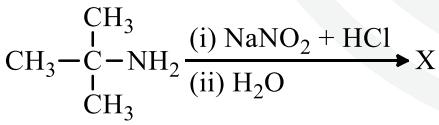
67. The major product (A) in the following reaction sequence is:



(1) $PhNH_2$ (2) $PhCONH_2$

(3) PhCN (4) PhNC

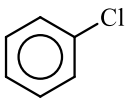
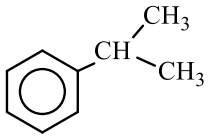
68. Major product (X) of the following reaction is:

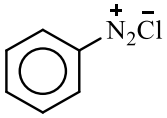
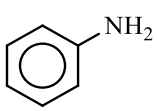


(1) Diazonium salt (2) 1°-Alcohol

(3) 2°-Alcohol (4) 3°-Alcohol

69. Most of the worldwide production of phenol is from:

(1)  (2) 

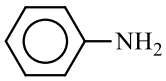
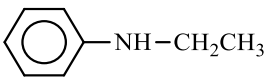
(3)  (4) 

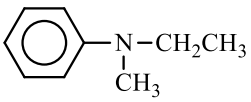
70. Sodium benzoate on heating with sodalime gives majorly:

(1) C_6H_6 (2) C_6H_5OH

(3) C_6H_{12} (4) C_2H_4

71. Which of the following amines can be prepared by Gabriel phthalimide reaction?

(1)  (2) 

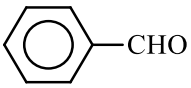
(3) $CH_3CH_2NH_2$ (4) 

72. Isobutyl alcohol and n-butyl alcohol are related as:

(1) Chain isomers (2) Position isomers

(3) Metamers (4) stereo isomers

73. Tollen's test is **not** given by:

(1) HCHO (2) 

(3) CH_3COCH_3 (4) HCOOH

74. The activating effect of $–NH_2$ group during its ArS_E reaction can be controlled by its:

(1) sulphonation (2) nitration

(3) acetylation (4) bromination

75. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

Assertion A: Greater the number of methyl groups attached to a negatively charged carbon atom, more will be the stability of carbanion.

Reason R: More the methyl group, more will be +I effect.

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

76. Given below are two statements:

Statement I: 2,4-DNP derivatives of carbonyl compounds are yellow, orange or red solids.

Statement II: Chemically Schiff's base is oxime.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.

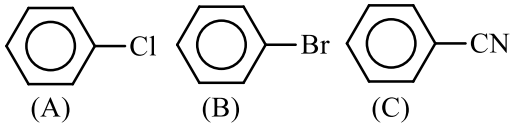
(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

[Contd...

77. Which among the following compounds can be prepared by Sandmeyer's reaction?



(A) (B) (C)

(1) A and B only (2) B and C only
(3) A and C only (4) A, B and C

78. Which among the following compounds is most reactive towards S_N2 reaction?
(1) CH_2=CH–Cl (2) $(CH_3)_2CHCl$
(3) CH_3Cl (4) $(CH_3)_3CCl$

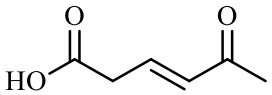
79. Maximum boiling point among the following is of:
(1) CH_3CHO (2) CH_3CH_2OH
(3) CH_3OCH_3 (4) $CH_3CH_2CH_3$

80. The structure of (C) in the given reactions is:

$$CH_3CH_2Br \xrightarrow{KCN} A \xrightarrow{H_2O_2+NaOH} B \xrightarrow{LiAlH_4} C$$

(1) $CH_3CH_2CH_2OH$ (2) $CH_3CH_2NHCH_3$
(3) $CH_3CH_2CH_2NH_2$ (4) $CH_3CH_2NH_2$

81. IUPAC name of given compound is:



(1) 1-Hydroxy hex-3-ene-1,5-dione
(2) 6-Hydroxy hex-3-ene-2,6-dione
(3) 5-Oxo hex-3-enoic acid
(4) 5-Oxo pent-3-en-1-carboxylic acid

82. Most suitable reactions/reagents for the following conversion is:

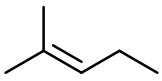
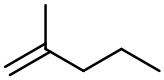
$CH_3CH=CH_2 \rightarrow CH_3CH_2CHO$

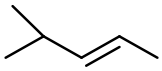
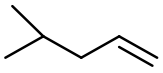
(1) (i) HBr, (ii) $KMnO_4/H^+$
(2) (i) B_2H_6, H_2O_2/NaOH, (ii) Cu + Δ
(3) (i) $KMnO_4/OH^-$, (ii) H_3O^+
(4) (i) HBr/peroxide, (ii) aq. KOH

83. Number of 1°-amines (including stereoisomers) corresponding to molecular formula $C_4H_{11}N$ is:
(1) 3 (2) 4
(3) 5 (4) 6

84. At 100°C, the vapour pressure of pure liquid 'A' is 500 mm of Hg and that of pure liquid 'B' is 800 mm of Hg. If a mixture of solutions A and B boils at 100°C and 760 mm of Hg pressure, the mole fraction of B in the mixture is:
(1) 0.63 (2) 0.75
(3) 0.87 (4) 0.92

85. Which of the following compounds on ozonolysis give acetone?

(1)  (2) 

(3)  (4) 

86. The **correct** order of basic strength for methyl substituted amines in aqueous solution is:
(1) 3° > 2° > 1° (2) 1° > 2° > 3°
(3) 2° > 1° > 3° (4) 2° > 3° > 1°

87. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Higher order (>3) reactions are rare.
Reason R: It is due to low probability of simultaneous collision of all the reacting species.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

88. Major product of the following reaction is:

$$\text{Cumene} \xrightarrow[(ii)\ H_3O^+]{(i)\ KMnO_4-KOH,\ \Delta}$$

(1) Benzoic acid
(2) Phenol
(3) Phenylacetic acid
(4) Benzoquinone

89. Most reactive halide with amine is:
(1) RI (2) RBr
(3) RCl (4) RF

90. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: For weak electrolytes, conductivity increases with dilution.
Reason R: Dissociation of weak electrolytes increases on dilution.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

[Contd...

91. Given below are two statements:
Statement I: Cytochrome *c* is a small protein attached to the inner surface of the inner mitochondrial membrane.
Statement II: Cytochrome *c* acts as a mobile carrier for transfer of electrons between complex I and II in ETS.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

92. Identify the **correct** feature of the organism given below:



(1) They produce zoospores endogenously.
(2) They are used extensively in biochemical and genetic work.
(3) They do not possess sex organs.
(4) During sexual reproduction, plasmogamy is immediately followed by karyogamy.

93. What is the respiratory quotient (RQ) of glucose?
(1) 0.9 (2) 1.0
(3) 0.7 (4) 1.4

94. Plants can get along without specialized respiratory organs because:
(1) transport of gases is very large.
(2) each plant part takes care of its own respiratory needs.
(3) roots respire at rates higher than animals.
(4) availability of O_2 is always a problem.

95. The margin of the thalamus grows upward enclosing the ovary completely and getting fused with it in:
(1) china rose. (2) plum.
(3) cucumber. (4) mango.

96. Given below are two statements:
Statement I: The F_1 headpiece of ATP synthase is an integral membrane protein complex.
Statement II: The passage of protons through the F_0 channel of ATP synthase is coupled to the catalytic site of the F_1 component for ATP production.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

97. In the plasma membrane, the quasi-fluid nature of lipids enables:
(1) movement of non-polar molecules across the membrane.
(2) lateral movement of proteins within the bilayer.
(3) movement of water molecules against the concentration gradient.
(4) passive movement of all ions across the membrane.

98. The complete combustion of glucose produces end products as:
(1) CO_2 and ethanol.
(2) CO_2, H_2O, and energy.
(3) lactic acid and energy.
(4) pyruvate and ATP.

99. Which of the following condition would result in inbreeding depression?
(1) Pollen release and stigma receptivity are not synchronized.
(2) Anther and stigma are placed at different positions.
(3) Self-incompatibility.
(4) Production of cleistogamous flowers.

100. How many protons are pumped by Complex II during electron transport in the ETS?
(1) Two (2) Four
(3) Three (4) Zero

101. Centriole is composed of microtubules as:
(1) 9 (triplets) + 2 (singlets) array.
(2) 9 (triplets) + 0 array.
(3) 9 (doublets) +2 (singlets) array.
(4) 9 (doublets) + 0 array.

102. The conversion of glucose-6-phosphate to fructose-6-phosphate is an example of:
(1) phosphorylation reaction.
(2) decarboxylation reaction.
(3) oxidation–reduction reaction.
(4) isomerisation reaction.

103. Which metabolite is common to the breakdown of fats, carbohydrates, and proteins before entering the Krebs' cycle?
(1) Pyruvic Acid
(2) Acetyl CoA
(3) PGAL
(4) DHAP

104. The sporophyte of *Marchantia*:
(1) is developed from the spore.
(2) is differentiated into a foot, seta, and capsule.
(3) is free-living and photosynthetic.
(4) produces sex organs at its apex.

105. How many ATP molecules are consumed during the formation of fructose-1,6-bisphosphate from one molecule of glucose during glycolysis?
(1) One
(2) Two
(3) Four
(4) Zero

106. Match **List-I** with **List-II**.

List-I (Enzyme)		**List-II (Catalysed reaction)**	
(A)	Lactate dehydrogenase	(I)	Acetyl-CoA + OAA → Citric acid
(B)	Invertase	(II)	Pyruvate → Lactic acid
(C)	Citrate synthase	(III)	Sucrose → Glucose + Fructose
(D)	Pyruvate dehydrogenase	(IV)	Pyruvate → Acetyl-CoA

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-III, B-II, C-IV, D-I
(3) A-I, B-IV, C-III, D-II
(4) A-II, B-IV, C-I, D-III

107. Which of the following tissues constitutes the stele in a dicot root?
(1) Endodermis + Pericycle + Vascular Bundles
(2) Pericycle + Vascular Bundles + Pith
(3) Cortex + Pericycle + Vascular Bundles
(4) Epidermis + Cortex + Endodermis

108. When the organism needs to synthesize fatty acids, the molecule is withdrawn from the respiratory pathway is:
(1) pyruvate.
(2) citric acid.
(3) acetyl CoA.
(4) succinyl CoA.

109. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): The respiratory pathway is considered an amphibolic pathway rather than a catabolic one.
Reason (R): The respiratory pathway involves both anabolism and catabolism.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

110. Which of the following events can RNA polymerase perform by itself, without association with any additional factors?
(1) Initiation of transcription
(2) Replication of the DNA template
(3) Elongation of transcription
(4) Termination of transcription

111. The final hydrogen acceptor in the electron transport system is:
(1) cytochrome a_3.
(2) oxygen.
(3) water.
(4) ubiquinone.

112. Each bivalent of chromosomes is clearly visible as a tetrad during:
(1) leptotene.
(2) zygotene.
(3) pachytene.
(4) anaphase II.

113. Identify the **correct** sequence of electron carriers in the Electron Transport System (ETS):
I. Cytochrome *b*
II. Ubiquinone
III. Cytochrome c_1
IV. Cytochrome *c*
V. Cytochrome $a - a_3$
Choose the **correct** answer from the options given below
(1) II → I → III → IV → V
(2) I → II → III → IV → V
(3) II → III → I → V → IV
(4) IV → I → II → III → V

114. The carbon skeleton produced during respiration is used as:
(1) energy source for photosynthesis.
(2) precursors for biosynthesis of other molecules.
(3) end product for excretion.
(4) substrate for fermentation only.

115. Identify the **correct** sequence of events occurring in the life cycle of a gymnosperm.
A. Pollen tube carrying the male gametes grows towards the archegonia
B. Formation of megasporangium
C. Production of seed
D. Development of the female gametophyte from the functional megaspore
E. Reductional division in one of the cells of the ovule
Choose the **correct** answer from the options given below:
(1) B → E → D → A → C
(2) E → B → D → A → C
(3) B → C → A → D → E
(4) A → C → B → D → E

116. In fermentation, what percentage of energy stored in glucose is released?
(1) Less than 7%
(2) About 13%
(3) About 90%
(4) Approximately 50%

117. For each ATP produced, how many proton(s) pass (es) through inter membrane channel of ATP synthase during mitochondrial oxidative phosphorylation?
(1) $2H^+$ (2) $1H^+$
(3) $4H^+$ (4) $10H^+$

118. In a typical angiosperm anther, the tapetum:
(1) nourishes the developing microspore mother cells.
(2) helps in the dehiscence of the anther.
(3) generally, has a single, large nucleus.
(4) forms the outermost wall layer of the microsporangium.

119. Net gain of NADH + H^+ during alcoholic fermentation is:
(1) two.
(2) three.
(3) one.
(4) zero.

120. Which of the following statements is **correct** about metabolic reactions?
(1) They are living things because they involve biomolecules.
(2) They are non-living things because they lack cellular organization.
(3) They are neither living nor non-living things, but are surely living reactions.
(4) They are non-living reactions as they do not occur inside a body.

121. Which of the following intermediates of the TCA cycle is formed after the first oxidative decarboxylation?
(1) Citric acid (2) α-Ketoglutaric acid
(3) Succinyl-CoA (4) Oxaloacetic acid

122. Given below are two statements:
Statement I: The energy released by oxidation in respiration is used directly by the cell for its metabolic needs.
Statement II: The energy produced during respiration is trapped as chemical energy in the form of ATP, which is used as the energy currency.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

123. Androecium in Leguminosae is represented by:
(1) A_{2+4}. (2) $A_{(5)}$.
(3) $A_{(9)+1}$. (4) A_5.

124. Given below are two statements:
Statement I: In aerobic respiration, there is a net gain of only 18 ATP for each molecule of glucose degraded.
Statement II: NADH is oxidized to NAD^+ very vigorously in fermentation compared to aerobic respiration.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

125. If one molecule of NADH undergoes oxidative phosphorylation in the electron transport system (ETS), how many ATP molecules will be produced?
(1) Two (2) One
(3) Three (4) Four

126. The genetic code is characterized by all, **except**:
(1) Triplet nature
(2) Universal without any exception
(3) Degenerate
(4) Presence of three stop codons and one start codon

127. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): Glycolysis occurs in aerobic organisms, facultative anaerobes, and obligate anaerobes.
Reason (R): Glycolysis is an oxygen-independent pathway.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

128. During the conversion of succinyl-CoA to succinic acid in the TCA cycle, one molecule of:
(1) NADH is produced.
(2) $FADH_2$ is produced.
(3) GTP is produced.
(4) CO_2 is released.

129. During glycolysis, which specific step involves the formation of NADH + H^+?
(1) 3 PGA → 2 PGA
(2) PGAL → BPGA
(3) PEP → Pyruvate
(4) 3 PGA → DHAP

130. In the mitochondrial Electron Transport System (ETS), which complex contains cytochromes *a* and a_3 and two copper centers?
(1) Complex V
(2) Complex III
(3) Complex IV
(4) Complex II

131. Match **List-I** with **List-II.**

List-I (Compound)		**List-II (Number of Carbons)**	
(A)	Lactic acid	(I)	4 carbons
(B)	Isocitrate	(II)	5 carbons
(C)	Fumarate	(III)	6 carbons
(D)	α-Ketoglutaric acid	(IV)	3 carbons

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-I, D-IV
(2) A-I, B-IV, C-III, D-II
(3) A-II, B-I, C-IV, D-III
(4) A-IV, B-III, C-I, D-II

132. Which of the following statements about glycolysis are **correct**?
A. It occurs in the mitochondrial matrix.
B. Glucose undergoes partial oxidation.
C. It involves a chain of 10 enzymatic reactions.
D. The end product is acetyl CoA.
Choose the **correct** answer from the options given below:
(1) A and B
(2) B and C
(3) C and D
(4) A and D

133. Glycerol enters the respiratory pathway after being converted to:
(1) Pyruvic acid
(2) Acetyl-CoA
(3) 3-phosphoglyceraldehyde
(4) 3-phosphoglycerate

134. When protein is used as a respiratory substrate:
(1) Volume of CO_2 consumed > Volume of O_2 evolved
(2) Volume of CO_2 evolved < Volume of O_2 consumed
(3) Volume of CO_2 evolved = Volume of O_2 consumed
(4) Volume of CO_2 consumed = Volume of O_2 evolved

135. Complex I of mitochondrial ETS is also known as:
(1) cytochrome *c* oxidase.
(2) cytochrome bc_1 complex.
(3) succinate dehydrogenase.
(4) NADH dehydrogenase.

136. Which of the following is a **correct** statement regarding evolution?

(1) Existing life forms share similarities only with modern organisms.

(2) The geological history of earth does not correlate with the biological history.

(3) Adaptive ability is inherited and has a genetic basis.

(4) Life appeared 4.5 billion years ago.

137. Which of the following statements is **incorrect**?

(1) Cancerous cells lose the property of contact inhibition, leading to the formation of tumors.

(2) Metastasis is the most feared property of malignant tumors

(3) Computed Tomography (CT) uses strong magnetic fields and non-ionising radiations to detect pathological changes.

(4) Several genes called cellular oncogenes have been identified in normal cells which, when activated, can lead to oncogenic transformation.

138. Match **List-I** with **List-II**.

List-I (Hominid)		**List-II (Feature)**	
(A)	*Homo habilis*	(I)	Brain size 900cc
(B)	*Homo erectus*	(II)	Brain size 1400cc; buried dead
(C)	*Neanderthal*	(III)	Brain size 650-800cc
(D)	*Australopithecines*	(IV)	Lived in East African grasslands; ate fruit

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV

(2) A-III, B-II, C-I, D-IV

(3) A-III, B-I, C-II, D-IV

(4) A-III, B-I, C-IV, D-II

139. Which of the following statements regarding Diaphragms, Cervical caps, and Vaults is **incorrect**?

(1) They are made of rubber and are inserted into the female reproductive tract to cover the cervix.

(2) Unlike condoms, these barrier methods are reusable.

(3) They are highly effective in preventing the transmission of STIs and AIDS.

(4) Spermicidal creams, jellies, and foams are usually used along with them to increase their contraceptive efficiency.

140. Identify the **correct** statements regarding the origin of life:

A. Universe is almost 13.8 billion years old.

B. Earth was formed 4.5 million years ago.

C. Life appeared 500 million years after the formation of Earth.

D. There was no atmosphere on early earth.

E. Stellar distances are measured in kilometers.

(1) A, B, C and D only

(2) B, C and D only

(3) A, D and E only

(4) A, C and D only

141. Which of the following is **correct** regarding the nature of enzyme action and its properties?

(1) With the increase in substrate concentration, the velocity of the enzymatic reaction decreases initially.

(2) Enzymes lower the activation energy, thereby increasing the rate of reaction.

(3) Most enzymes are made of carbohydrates and remain functional at very high temperatures.

(4) Enzyme catalysed reactions are slower than non enzyme catalysed reaction.

142. Paleontological evidence for evolution comes from the study of:

(1) Embryos

(2) Proteins

(3) Fossils

(4) Anatomy

143. Select the organism that is triploblastic, bilaterally symmetrical, and truly coelomate:

(1) *Taenia* (Tapeworm)

(2) *Ascaris* (Roundworm)

(3) *Pheretima* (Earthworm)

(4) *Adamsia* (Sea anemone)

[Contd...

144. H.M.S. Beagle was the sail ship used by:
(1) Alfred Wallace
(2) Charles Darwin
(3) Hugo de Vries
(4) Thomas Malthus

145. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Evolution is a stochastic process.
Reason R: Evolution is based on chance events in nature and chance mutations in organisms.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

146. Adaptive radiation in Australian marsupials and placental mammals is an example of:
(1) Divergent evolution
(2) Convergent evolution
(3) Saltation
(4) Spontaneous generation

147. The ejaculatory duct in human males is formed by the union of:
(1) Vas deferens and epididymis
(2) Prostate gland and urethra
(3) Seminiferous tubules and rete testis
(4) Vas deferens and duct of seminal vesicle

148. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Mutation Theory	(I)	Louis Pasteur
(B)	Life originated from pre-existing life	(II)	Hugo de Vries
(C)	Use and Disuse of organs	(III)	Alfred Wallace
(D)	Malay Archipelago	(IV)	Lamarck

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-IV, D-III
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-I, C-II, D-III

149. The occurrence of industrial melanism in moths in England is an example of:
(1) Evolution by anthropogenic action
(2) Natural selection
(3) Directional selection
(4) All of these

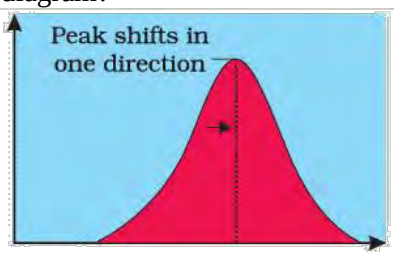
150. If a section of a population migrates to a new place and gene frequencies change by chance in the new sample, this phenomenon is **best** described as:
(1) Gene Flow
(2) Genetic Drift
(3) Natural Selection
(4) Saltation

151. Which of the following parts of the human brain is **correctly** paired with its primary function?
(1) Corpus callosum – Controls body temperature and urge for eating.
(2) Cerebellum – Regulates posture and maintains balance.
(3) Medulla oblongata – Acts as a relay station for sensory and motor signaling.
(4) Hypothalamus – Forms the outer layer of grey matter in the cerebrum.

152. Which of the following is **correct** statement regarding the evolution of man?
(1) *Homo habilis* probably ate meat.
(2) *Homo erectus* appeared about 3.5 million years ago.
(3) Neanderthals lived in near east and central Asia between 1,00,000-40,000 years back.
(4) *Dryopithecus* walked upright and was 4 feet tall and more man-like.

153. Which of the following hormones is synthesized by the hypothalamus but is stored and released by the neurohypophysis?
(1) Prolactin and TSH
(2) ACTH and MSH
(3) Growth Hormone and LH
(4) Oxytocin and Vasopressin

154. Which type of natural selection is represented in the given diagram?



(1) Stabilizing selection
(2) Disruptive selection
(3) Directional selection
(4) Genetic drift

[Contd...

155. The part of the nephron that is impermeable to water but allows active or passive transport of electrolytes is the:

(1) Proximal Convoluted Tubule
(2) Descending limb of loop of Henle
(3) Ascending limb of loop of Henle
(4) Collecting duct

156. Analysis of _____ content revealed compounds similar to Miller's experiment. Choose the option **correctly** filling the blanks.

(1) Air
(2) Meteorite
(3) Volcanic ash
(4) Lunar soil

157. Consider the following statements:

Statement I: Mutations are random and directionless.

Statement II: Darwinian variations are small and directional.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

158. Match **List-I** with **List-II**.

List-I (Timeframe)		**List-II (Event)**	
(A)	2000 mya	(I)	Invertebrates formed
(B)	500 mya	(II)	First cellular forms of life
(C)	350 mya	(III)	Sea weeds and few plants
(D)	320 mya	(IV)	Jawless fish evolved

Choose the **correct** answer from the options given below:

(1) A-I, B-III, C-IV, D-II
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-IV, C-II, D-I

159. Which of the following statements about Natural Selection are **correct**?

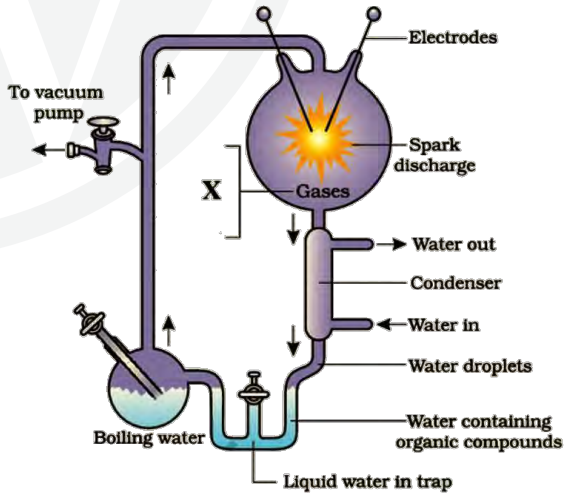
A. It is based on factual observations like limited natural resources.
B. Populations are generally stable in size except seasonal fluctuations.
C. Darwin was influenced by the work of Thomas Malthus on populations.
D. Alfred Wallace was co-proposer of theory of evolution by use and disuse of organs.
E. Reproductive fitness is the end result of the ability to adapt.

(1) A, B, C and E only
(2) A, C, D and E only
(3) B, C and D only
(4) All are correct

160. Identify the **correct** match between the joint and its location in the human body:

(1) Gliding joint – Between carpal and metacarpal of thumb.
(2) Hinge joint – Between the Atlas and Axis vertebrae.
(3) Pivot joint – Between humerus and pectoral girdle.
(4) Saddle joint – Between the carpals and metacarpals of the thumb.

161. In the given diagram labelled part **X** and the specific conditions used in closed flask is represented as:



(1) O_2, CH_4, H_2O, NH_3 at 800 ^oC
(2) CH_4, H_2, NH_3 and water vapor at 800 ^oC
(3) CH_4, H_2, NH_3 and water vapor at 100 ^oC
(4) CO_2, CH_4, H_2, NH_3 at 800 ^oC

162. Which of the following factors is **not** favorable for the formation of oxyhaemoglobin at the alveolar site?

(1) High pO_2
(2) Low pCO_2
(3) Lesser H^+ concentration
(4) Higher temperature

[Contd...

163. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Sweet potato and Potato	(I)	Homology
(B)	Lichens	(II)	Anthropogenic evolution
(C)	Herbicide resistance	(III)	Analogy (Root/Stem)
(D)	Vertebrate Hearts	(IV)	Do not grow in polluted areas

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-II, D-I
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-III, C-I, D-IV
(4) A-II, B-I, C-IV, D-III

164. According to the theory of special creation found in conventional religious literature, which of the following is **not** a correct connotation?

(1) All living organisms that we see today were created as such.
(2) The diversity of life forms has changed significantly since the time of creation.
(3) The earth is approximately 4000 years old.
(4) The diversity of life will remain the same in the future as it was at creation.

165. Choose the **correct** statements regarding evidences of evolution:

A. Fossils are remains of soft parts of life-forms found in rocks.
B. Embryological support was proposed by Ernst Heckel and disapproved by Karl Ernst von Baer.
C. Vestigial gill slits are found in all adult vertebrates.
D. Homology indicates common ancestry.
E. Biochemical similarities point to shared ancestry.

(1) C, D and E only (2) A, B and D only
(3) B, D and E only (4) A, B, C, D and E

166. Match **List-I** with **List-II.**

List-I (Placental mammal)		List-II (Australian marsupial)	
(A)	Bobcat	(I)	Spotted cuscus
(B)	Lemur	(II)	Numbat
(C)	Anteater	(III)	Flying phalanger
(D)	Flying squirrel	(IV)	Tasmanian tiger cat

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-I, C-III, D-II
(3) A-III, B-IV, C-I, D-II
(4) A-IV, B-I, C-II, D-III

167. Heroin, commonly called 'smack', is chemically _____ and is synthesized by the _____ of morphine. Choose the option **correctly** filling the blanks.

(1) Diacetylmorphine, acetylation
(2) Diacetylmorphine, methylation
(3) Benzodiazepine, acetylation
(4) Cannabinoid, alkylation

168. Which of the following groups of Sexually Transmitted Infections (STIs) are **not** completely curable even if detected early?

(1) Syphilis, Gonorrhoea, Genital warts
(2) Chlamydiasis, Trichomoniasis, Syphilis
(3) Genital herpes, Hepatitis-B, HIV infection
(4) Genital warts, Gonorrhoea, Chlamydiasis

169. Identify the **correct** statements regarding the evolution of plants and animals:

A. The first organisms to invade land were animals.
B. Coelacanth was caught in South Africa in 1938.
C. *Tyrannosaurus rex* was about 20 feet in height.
D. Dinosaurs disappeared suddenly about 65 million years ago.
E. Small-sized reptiles of the dinosaur era still exist today.

(1) B, C, D and E only
(2) A, B, C and D only
(3) C, D and E only
(4) A, B, C, D and E

170. Oparin and Haldane proposed that life was preceded by:

(1) Biological evolution
(2) Chemical evolution
(3) Spores from Universe
(4) Anthropogenic evolution

171. Given below are two statements:

Statement I: The first non-cellular forms of life could have been giant molecules like RNA, Protein, and Polysaccharides.

Statement II: The first non-cellular forms of life did not originate till about 2000 million years ago.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

172. Regarding the Hardy-Weinberg Principle, which of the following statements are **true**?

A. Gene pool remains constant.
B. Sum total of all allelic frequencies is 1.
C. Genetic drift occurs due to gene migration multiple times.
D. Only two factors genetic drift and gene flow affect Hardy-Weinberg Principle.
E. Natural selection leads to stabilization if more individuals acquire mean character value.

(1) A, B, D and E only
(2) A, B, C and D only
(3) B, D and E only
(4) A, B and E only

.

173. The idea of 'Panspermia' suggests life units called _____ were transferred between planets.

Choose the option **correctly** filling the blank.

(1) Cysts (2) Seeds
(3) Spores (4) Gametes

174. In a standard ECG, the 'T-wave' represents:

(1) Depolarisation of the atria
(2) Depolarisation of the ventricles
(3) Return of the ventricles from excited to normal state (repolarisation)
(4) End of the atrial systole

175. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Mesozoic Era	(I)	Lower invertebrates
(B)	Proterozoic Era	(II)	Fish and Amphibia
(C)	Cenozoic Era	(III)	Birds and Reptiles
(D)	Paleozoic Era	(IV)	Mammals

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-IV, D-III
(2) A-III, B-I, C-IV, D-II
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-I, C-II, D-IV

176. Select the **correctly** matched pair regarding human evolution:

(1) *Ramapithecus* — More ape-like
(2) *Dryopithecus* — More man-like
(3) Agriculture — 10,000 years ago
(4) Pre-historic cave art — 1,00,000 years ago

177. In frogs, the process of respiration during aestivation and hibernation takes place through which of the following?

(1) Lungs only
(2) Buccal cavity and lungs
(3) Skin
(4) Gills

178. Select the **correctly** matched pair.

(1) Darwin's Finches — Convergent Evolution
(2) Branching Descent — Key concept of Darwinian Theory
(3) Saltation — Small and directional variations
(4) Louis Pasteur — Gave theory of Spontaneous Generation

179. Which of the following is a type of loose connective tissue that serves as a support framework for epithelium and contains fibroblasts, macrophages, and mast cells?

(1) Areolar tissue (2) Adipose tissue
(3) Tendons (4) Cartilage

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The skull of a baby chimpanzee is more like an adult human skull than an adult chimpanzee skull.

Reason R: Mammals are viviparous and protect their unborn young inside the mother's body.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

■■■

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